

TrES-5b

R-0765

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_231103 · created 2026-07-28T04:26:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460251.6675 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.027
Transit depth (Rp/Rs)^2	3.38 +/- 1.0 [%]
Orbital Inclination (inc)	82.31 +/- 2.32
Airmass coefficient 1 (a1)	1.13 +/- 0.03
Airmass coefficient 2 (a2)	-0.089 +/- 0.021
Scatter in the residuals of the lightcurve fit is	2.67 %
Best Comparison Star	None
Optimal Aperture	3.03
Optimal Annulus	6.26
Transit Duration (day)	0.059 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.027 vs 0.142 (deviation 1.51σ)
Duration fit vs published	0.059 d vs 0.0758 d (deviation 0.71σ)
Mid-time O–C	9.1 min
Depth SNR	6.6
Residual scatter	2.67 %

Light curve

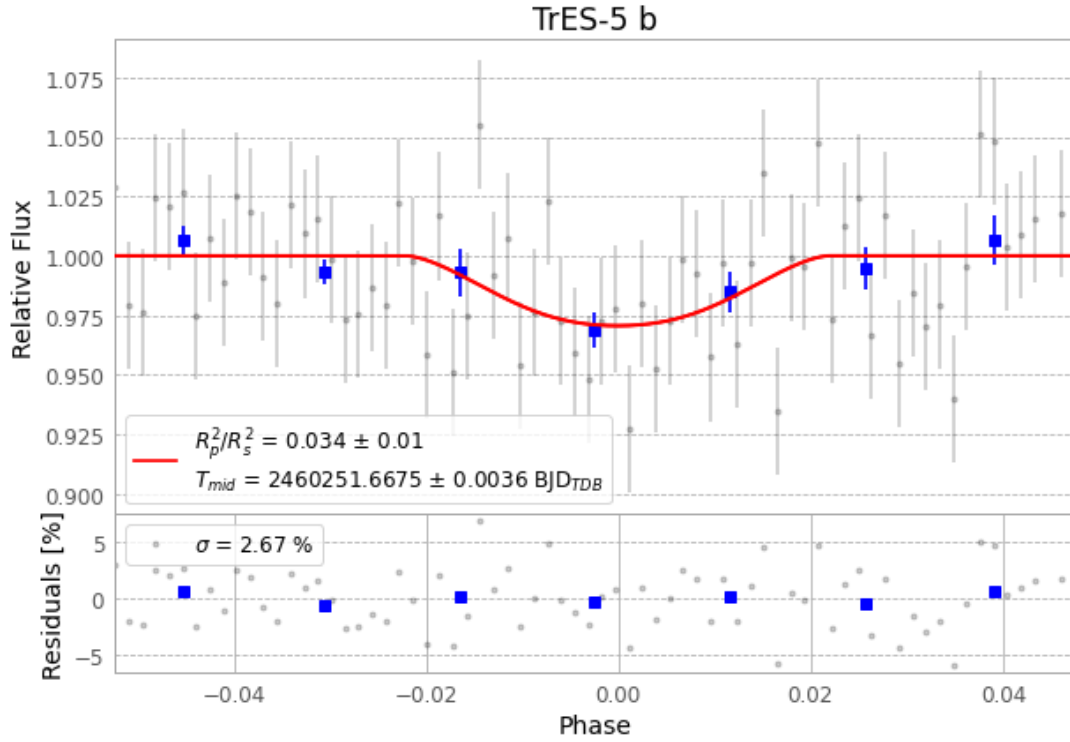


Figure 1 — FinalLightCurve_TrES-5 b_2023-11-02.png (SHA-256 4307227fa408c895...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 206], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0fbcd3a0fbd26089...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5231103020616.FITS → TRES-5231103053915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-231103.log (5922b740b6df...), FinalLightCurve_TrES-5 b_2023-11-02.png (4307227fa408...), FinalLightCurve_TrES-5 b_2023-11-02.csv (3075d20b44c5...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:26Z.